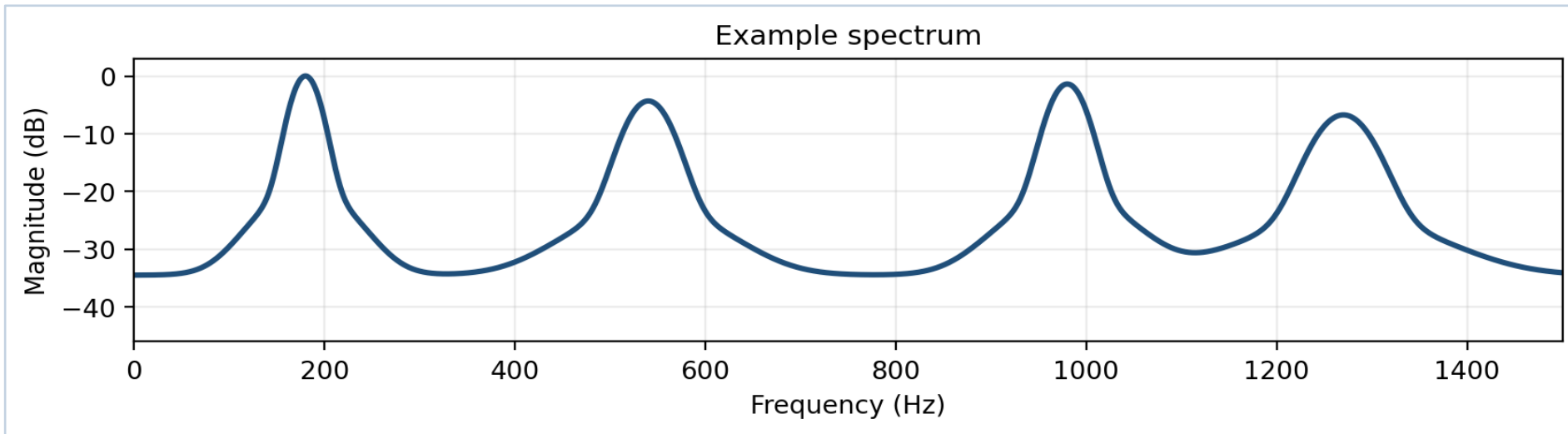


Topic 9

Computing the Spectrum

DFT • FFT • windowing • leakage • resolution • zero-padding • spectrograms



Key idea. A computer estimates spectra from finite sampled data; the DFT is the central computational tool.

From the Fourier integral to a finite Fourier sum

Replace the integral by a Riemann sum

$$X_c(j\omega) = \int x_c(t) e^{-j\omega t} dt$$

$$X_c(j\omega) \approx T_s \sum_{n=0}^{N-1} x_n e^{-j\omega n}$$

- Sample at $t=nT_s$.
- Keep only N measured samples.
- Replace dt by T_s .

Key idea. The finite sum approximates the CTFT at selected physical frequencies.

Choosing the DFT frequency grid

N equally spaced analysis frequencies

$$\omega_k = \frac{2\pi k}{N}, k = 0, \dots, N - 1$$

$$X_k = \sum_{n=0}^{N-1} x_n e^{\frac{-j2\pi kn}{N}}$$

- The DFT is the finite Fourier sum evaluated on a special equally spaced grid.
- $T_s X[k]$ approximates the CTFT at $\omega = \omega_k$.
- The DFT has an exact inverse for the N samples.

Key idea. The DFT is not a different Fourier idea; it is a discrete sampling of frequency for a finite record.

Example 1: a 4-point DFT

Problem. For $x[n]=\{1,1,0,0\}$, compute $X[k]$ directly for $k=0,1,2,3$.

1 $X_k = \sum_{n=0}^3 x_n e^{\frac{-j2\pi kn}{4}}$ Only $n=0$ and $n=1$ contribute.

2 $X_0 = 1 + 1 = 2$

3 $X_1 = 1 + e^{-j\frac{\pi}{2}} = 1 - j$; $X_2 = 1 + e^{-j\pi} = 0$

4 $X_3 = 1 + e^{-j\frac{3\pi}{2}} = 1 + j$ conjugate of $X[1]$

Final answer: $X = \{2, 1 - j, 0, 1 + j\}$

How the Fourier representations are related

Continuous/discrete time versus continuous/discrete frequency

	Continuous frequency	Discrete frequency
Continuous time	CTFT	Fourier series
Discrete time	DTFT	DFT

Key idea. All four use projection onto complex exponentials; the differences are signal and frequency assumptions.

DFT as samples of the DTFT

Keep three operations conceptually separate

$$X_k = X(e^{j\hat{\omega}}) \Big|_{\hat{\omega} = \frac{2\pi k}{N}}$$

- Sampling in time makes the DTFT periodic.
- Windowing changes the shape of the spectrum.
- The DFT samples that windowed DTFT on a discrete grid.

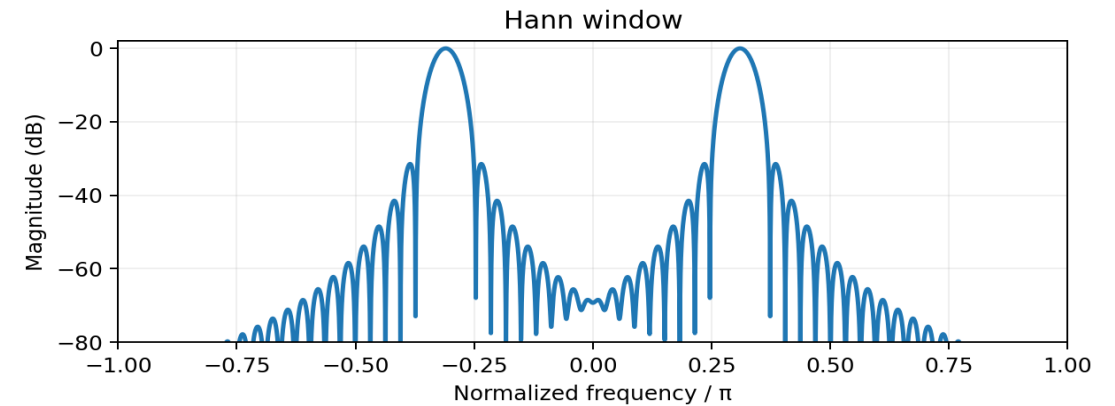
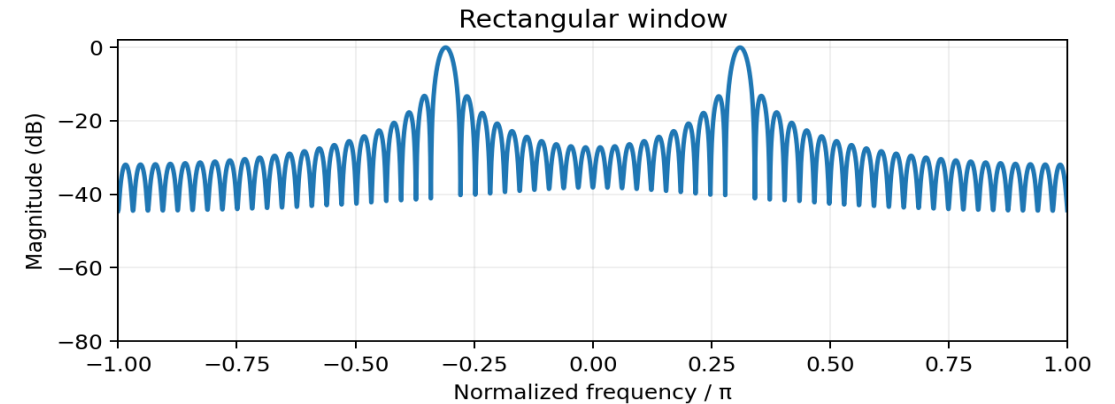
Key idea. Aliasing, leakage, and DFT-bin sampling are different phenomena.

Time windowing and spectral leakage

Finite observation spreads ideal spectral lines

$$x_{wn} = x_n w_n$$

- A DFT always operates on a finite record.
- Multiplication by a window in time corresponds to convolution in frequency.
- An ideal line becomes a shifted copy of the window spectrum.



Key idea. Spectral leakage is deterministic spreading caused by finite observation.

Rectangular versus Hamming/Hann windows

Mainlobe width versus sidelobe level

- Rectangular: narrower mainlobe → better nominal resolution.
- Rectangular: higher sidelobes → more leakage from strong components.
- Hamming/Hann: lower sidelobes → better weak-component visibility.
- Hamming/Hann: wider mainlobe → more data needed for the same resolution.

Key idea. Window choice is a trade-off between resolving close frequencies and suppressing leakage.

Example 2: choose a window

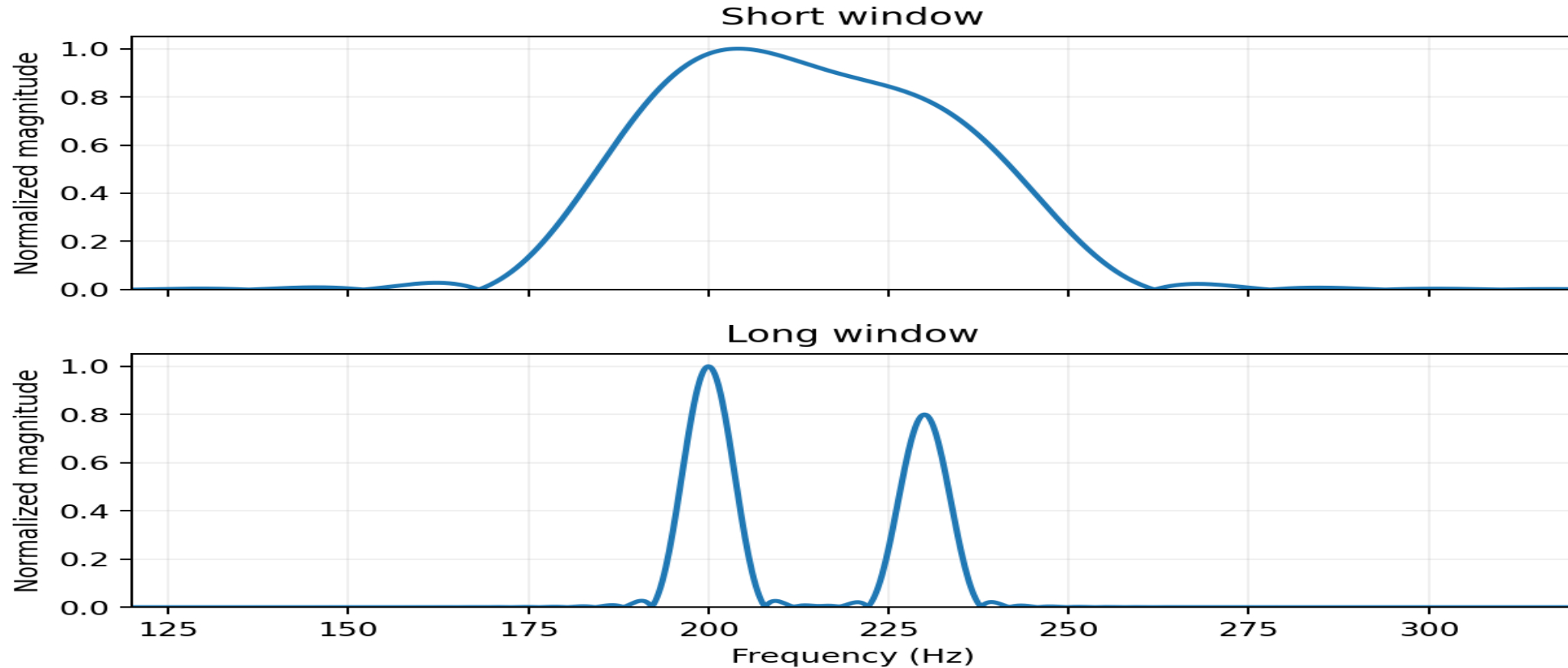
Problem. Two equal-amplitude tones at 1000 Hz and 1080 Hz are observed. Which window gives better separation for a fixed duration, and which better suppresses leakage?

- 1 The tones are 80 Hz apart.
- 2 For a fixed duration, the rectangular window has the narrower mainlobe.
- 3 Therefore rectangular gives better separation of comparable nearby tones.
- 4 Hamming/Hann has much lower sidelobes, so it is preferable when a weak tone sits near a strong one.

Final answer. Rectangular favors resolution; Hamming/Hann favors sidelobe suppression and dynamic range.

Resolution trade-off

Long windows sharpen frequency but blur time



Key idea. Only a longer nonzero observation improves true frequency resolution.

The DFT and inverse DFT

Exact analysis–synthesis pair for N samples

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}$$
$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}$$

- The forward DFT produces N complex coefficients.
- The IDFT reconstructs the original N samples exactly.
- For real x_n , $X_{N-k} = X^*[k]$.

Key idea. The DFT/IDFT pair is exact for the finite data record.

DFT bins and negative frequencies

Map bin indices to physical frequency correctly

$$\Delta_f = \frac{f_s}{N}, f_k = k \frac{f_s}{N}$$

Bins 0...N/2 represent DC through Nyquist for even N.

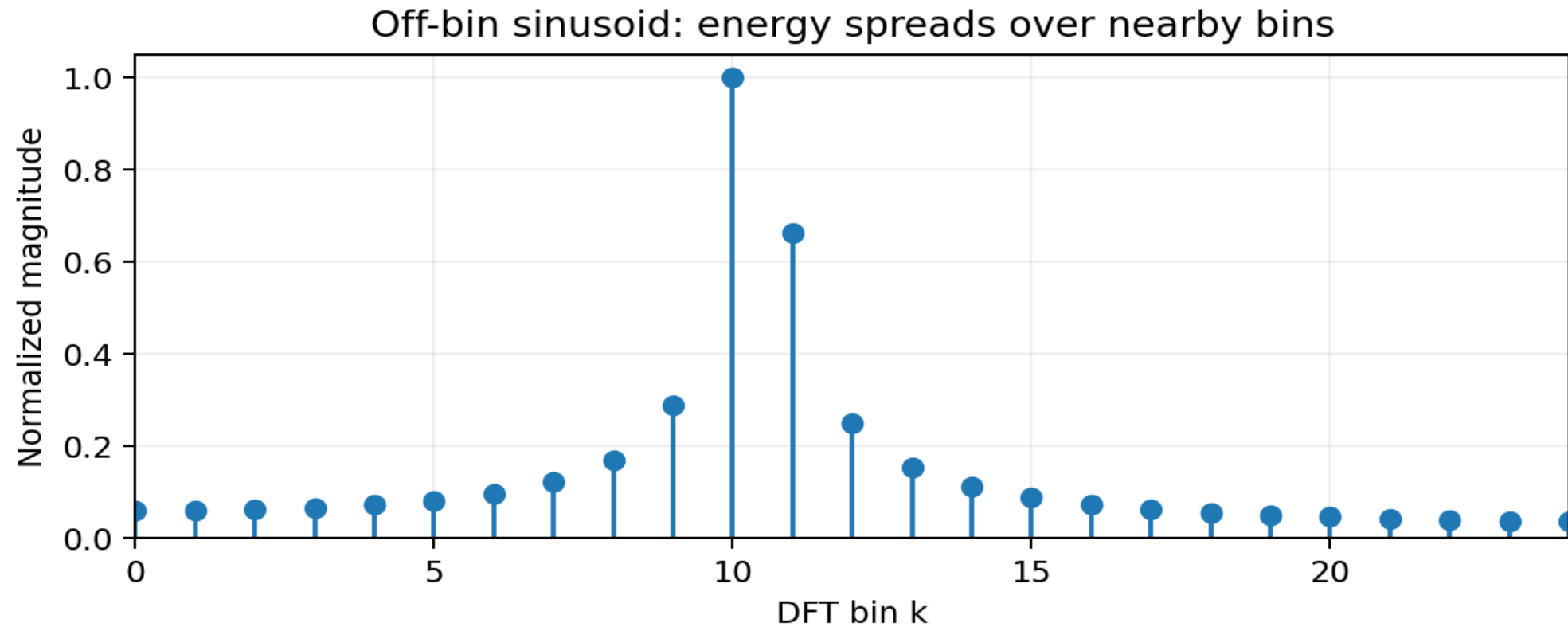
Upper-half bins represent negative frequencies: $f_k = (k-N)f_s/N$.

For real signals, the upper half is the conjugate partner of the lower half.

Key idea. Do not interpret every DFT bin as a new positive frequency.

Bin-centred and non-bin-centred sinusoids

The “picket-fence” effect



Key idea. An off-bin sinusoid spreads across DFT coefficients even when the FFT is perfectly correct.

Example 3: bin-centred or not?

Worked problem

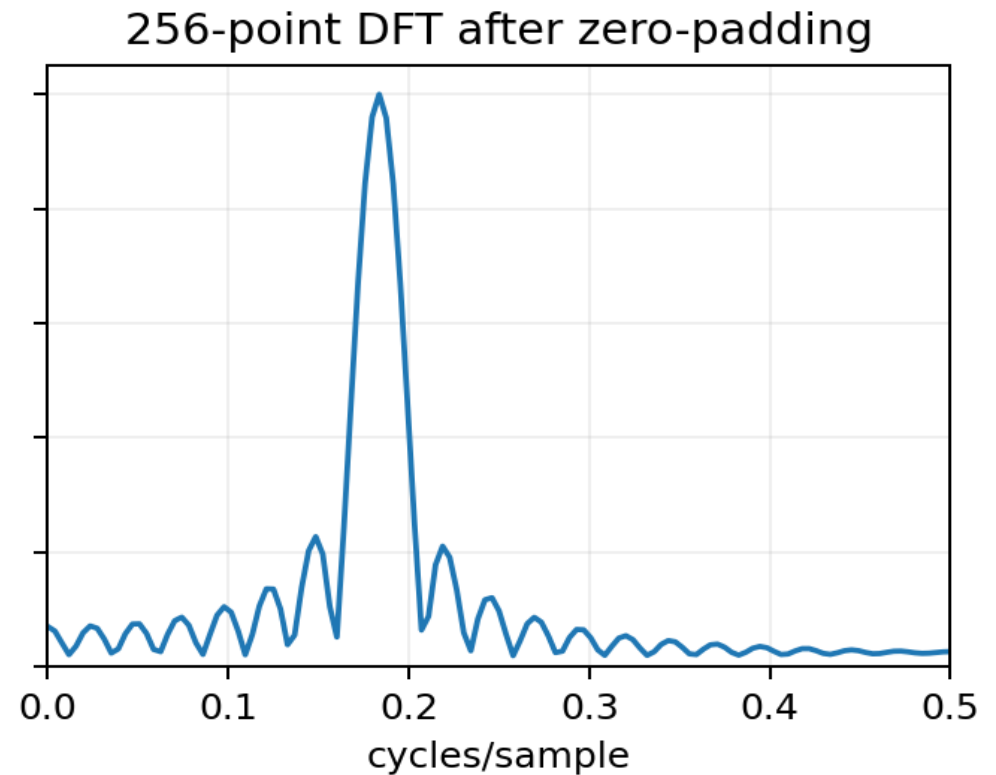
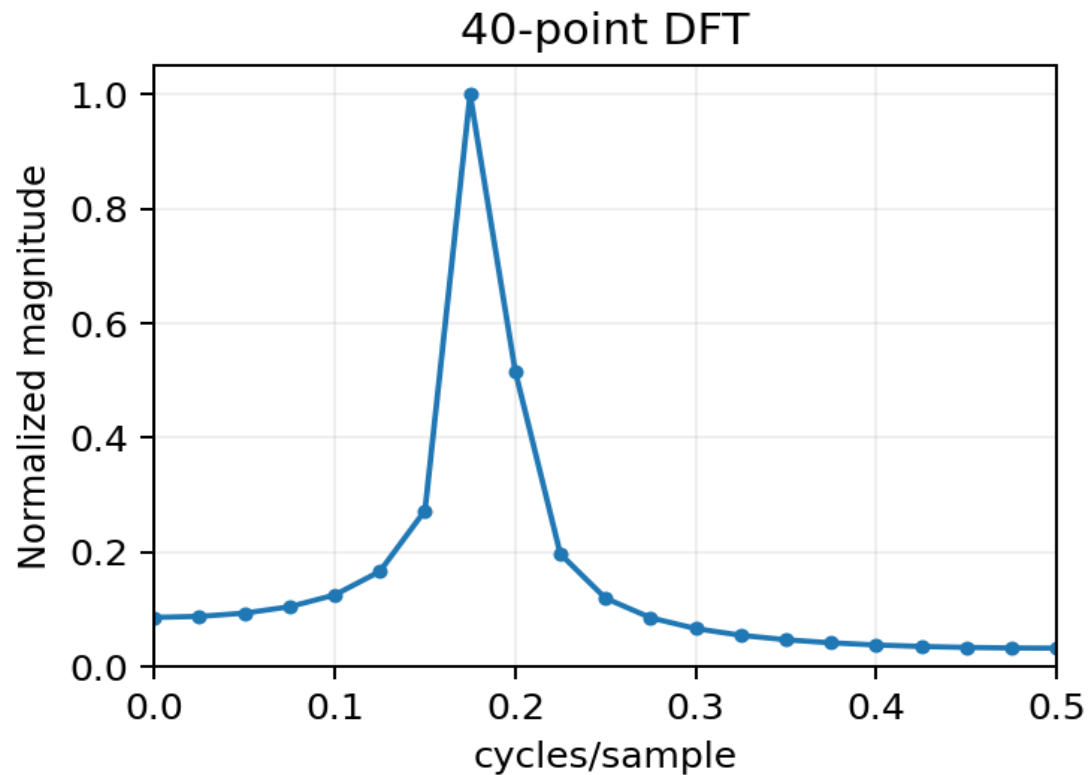
Problem. A signal is sampled at $f_s=1024$ Hz and analysed with $N=256$. Compare 128 Hz and 130 Hz.

- 1 Bin spacing is $\Delta f=1024/256=4$ Hz.
- 2 $128/4=32 \rightarrow$ exactly bin-centred at $k=32$.
- 3 $130/4=32.5 \rightarrow$ halfway between bins 32 and 33.
- 4 The 128-Hz cosine ideally occupies two conjugate bins; the 130-Hz cosine leaks into neighbours.

Final answer. 128 Hz is bin-centred; 130 Hz is not and exhibits leakage/picket-fence sampling.

Zero-padding

Denser frequency sampling without new information



Key idea. Zero-padding makes the spectral curve smoother to view but does not narrow the window mainlobe.

Example 4: what does zero-padding change?

Problem. A 40-sample sinusoid is analysed with a 40-point DFT and with a 256-point DFT after appending zeros.

- 1 The 256-point DFT has a denser frequency grid.
- 2 Both analyses contain the same 40 nonzero samples.
- 3 Therefore both have the same window duration and the same mainlobe width.
- 4 A true 256-sample observation would improve resolution because the physical record is longer.

Final answer. Zero-padding improves interpolation/visual density, not true resolving power.

Periodic signals and coherent sampling

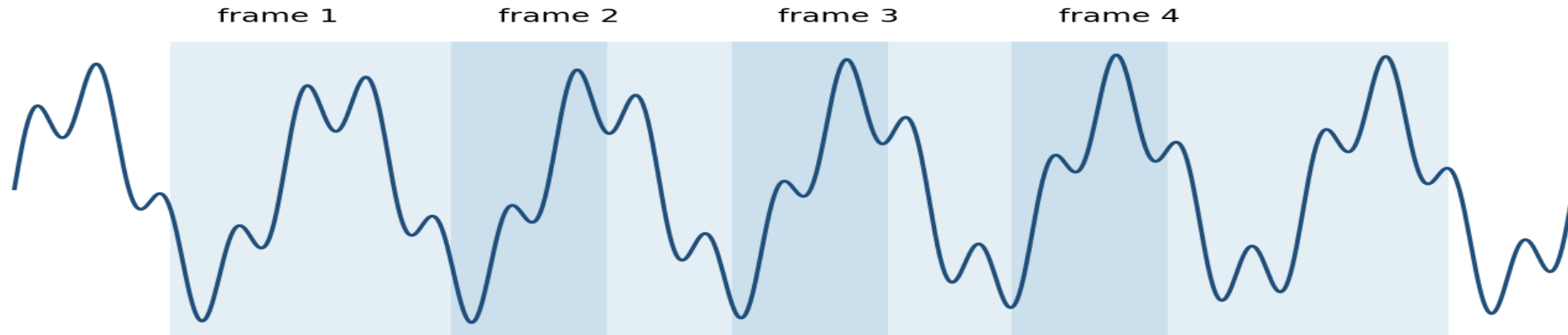
Choose an integer number of periods whenever possible

- If one complete discrete-time period has N samples, its N -point DFT gives the Fourier-series coefficients up to scale.
- Harmonics fall exactly on DFT bins.
- If the record cuts the periodic waveform at mismatched endpoints, the implied periodic extension has a discontinuity and leaks.

Key idea. Coherent sampling aligns the record boundaries and avoids artificial leakage.

The spectrogram

Spectrum as a function of time

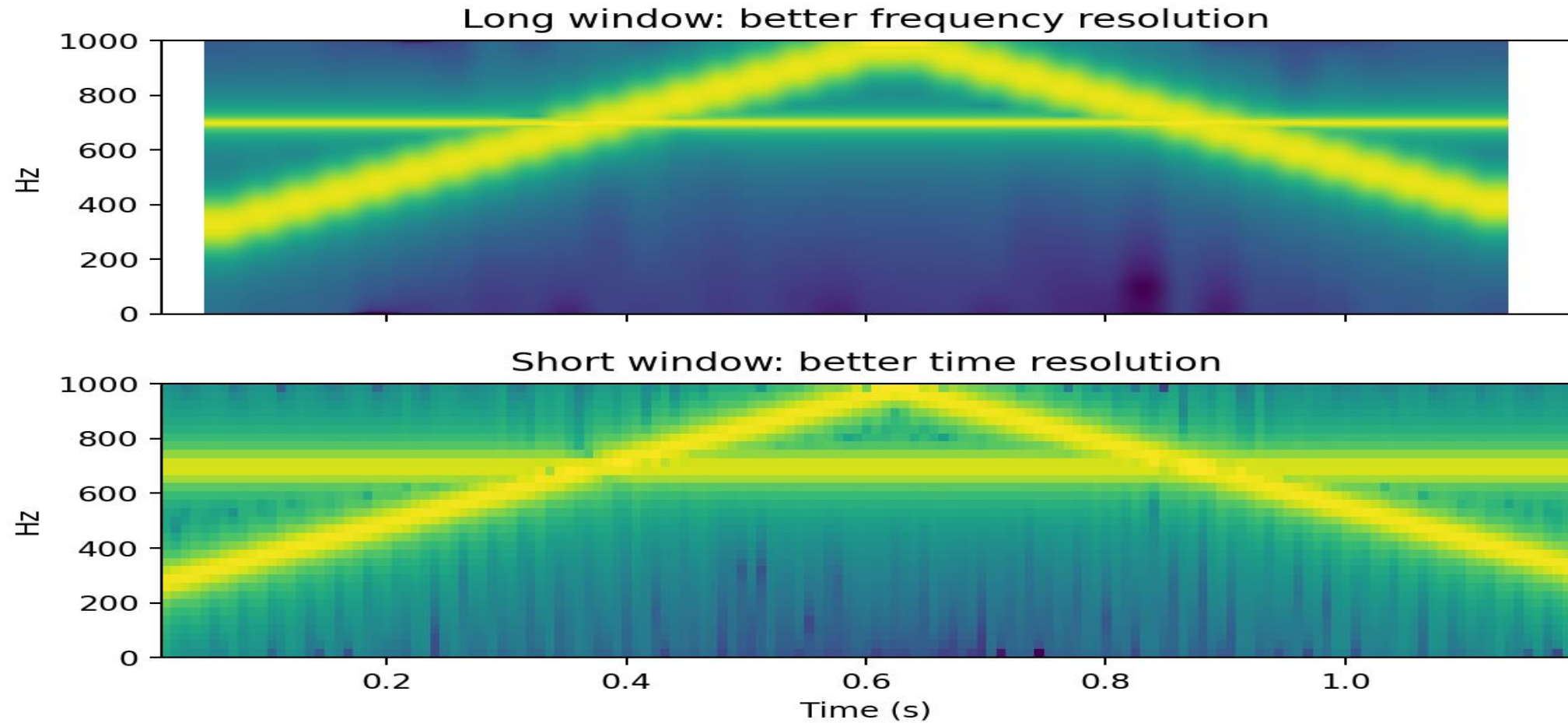


- Move a short window along the signal.
- Compute an FFT for each frame.
- Display magnitude (often in dB): time horizontally, frequency vertically.
- Frames commonly overlap by 50–80%.

Key idea. A spectrogram is a sequence of short-time spectra.

Spectrogram resolution

The same time–frequency trade-off appears again



Key idea. Long windows separate nearby frequencies; short windows localize rapid changes.

Example 5: choose a spectrogram window

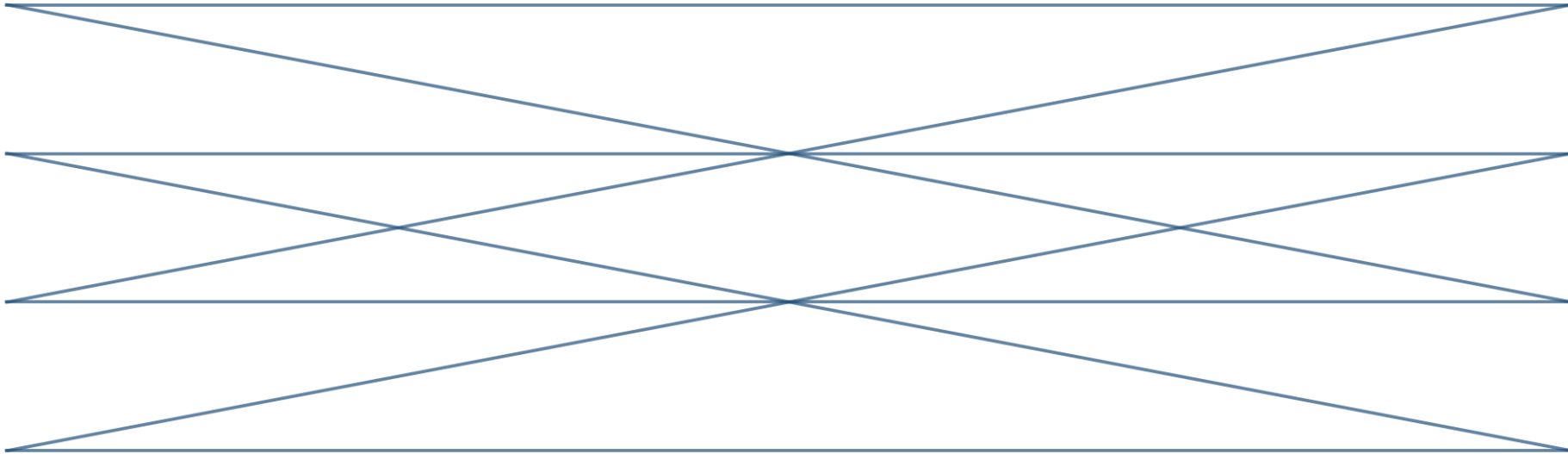
Problem. You need to distinguish two tones 25 Hz apart, but also locate note onsets within about 20 ms. What is the trade-off?

- 1 A longer window narrows spectral mainlobes and helps separate the 25-Hz spacing.
- 2 But the longer window averages over more time and smears onset timing.
- 3 A shorter window improves onset localization but may merge the two tones.
- 4 Choose the shortest window that still satisfies the needed frequency resolution, then set overlap/hop for smooth time tracking.

Final answer. There is no single window that simultaneously maximizes both time and frequency resolution.

Why the FFT is fast

Same DFT, fewer repeated calculations



- Direct DFT complexity: $O(N^2)$.
- FFT complexity: approximately $O(N \log_2 N)$.
- Radix-2 algorithms recursively split the data into even and odd indices.
- The FFT is an algorithm for computing the DFT, not a different transform.

Key idea. The FFT gains speed by reusing intermediate subtransform results.

Practical workflow for reliable spectrum analysis

A checklist before trusting an FFT plot

1. Check sampling rate and possible aliasing.
2. Choose a segment long enough for resolution but short enough for approximate stationarity.
3. Choose the window according to mainlobe/sidelobe needs.
4. Choose FFT length $N_{\text{fft}} \geq \text{record length}$; use zero-padding only for denser sampling.
5. Scale and label magnitude/power/PSD and frequency axes correctly.
6. Validate with known sinusoids and inspect bin alignment.

Key idea. A good spectrum is as much about measurement choices as about the FFT call itself.

Common interpretation errors

What the FFT plot does—and does not—mean

- FFT bins are samples of the finite, windowed signal spectrum—not the “true” continuous spectrum.
- Zero-padding does not improve true resolution.
- Spread peaks may be deterministic leakage, not noise.
- Windowing changes measured amplitudes; coherent-gain correction may be needed.
- Very long windows can hide temporal changes.

Key idea. Interpret the data model first: sampled, windowed, then evaluated on a frequency grid.

Notation and takeaways

Minimum language for computing the spectrum

N : number of nonzero samples / window length.

N_{fft} : DFT/FFT length; may exceed N through zero-padding.

$\Delta f = f_s / N_{\text{fft}}$: frequency-grid spacing.

$w[n]$: analysis window.

$X[k]$: DFT coefficient.

Hop size: frame shift used for the spectrogram.

Sampling can alias.

Windowing causes leakage.

Longer data improves true frequency resolution.

Zero-padding only samples more densely.

The FFT computes the DFT efficiently.

Repeated windowed FFTs form a spectrogram.